EXECUTION PROCEDURE — Team VEL PGK Proof-of-Concept

Scope: proof_of_concept.slx (Simulink 7-DoF dual-spin GNC PoC) + Monte-Carlo / CEP campaign.
CEP REFERENCE: All CEP / statistical accuracy numbers are taken from the Power Generation folder — Power Generation (Alternator). CEP is NOT recomputed from the Simulink tree.

1. Prerequisites

- **MATLAB:** R2023b+ with Simulink (no extra toolboxes for the PoC; Parallel Computing Toolbox optional for the MC fan-out).
- **Core Model Directory:** VEL_drive/ containing proof_of_concept.slx + live3d.m (same folder).
- **Power & CEP Repository:** Repo ~/projects/VEL/VEL-repo/pgk_power_sim (MC/CEP pipeline + campaign log).
- **Callbacks:** Model InitFcn callback auto-adds the model folder to the MATLAB path (resolves live3d); StopFcn auto-generates the 3-D trajectory figure at stop.

2. Pre-flight Checks (Simulink)

1. **Command Window Initialization:**
   IC0 = [0;0;-4.5;449.5;0;-404.8;0.93358;0;0.35837;0;1131;0;0;0;1131];
2. **Integrator Block:** Integrator (1/s) block initial condition field must be set to IC0 (literal name).
3. **Solver Settings:** ode4 (Runge-Kutta), fixed step 0.002 s; Stop Time = 120 s.
4. **Sensors Block Configuration:**
   - Sample time = 0.005 (IMU 200 Hz; GPS 50 Hz generated internally).
   - Inputs = (t, truth state 15) only — no dstate wiring (avoids algebraic loop).
5. **Ground Halt Chain:** Enabled via Demux port 3 (D) → Compare To Constant >= 0 → Stop Simulation.
6. **Execution Modes:** Demo runs keep Plot3D_Live active. For batch/long runs, comment it out to eliminate drawnow overhead.

3. Nominal Run & Verification

Press **Run** (~1–2 min). Expected event timeline: safe-arm 0.85 s → guidance arm/hold 7 s → ZEM nulled ≈44 s → ground halt ≈68 s (D ≥ 0 trigger) → StopFcn 3-D figure pops.

Verification Checklist

- **Navigation Accuracy:** Nav overlays truth (N, D within tens of metres).
- **zem_out Signal:** 0 → ~3.4 km arm bump → decay to ≈0 mid-descent (terminal predictor residual documented).
- **Actuation & Power:** F_can ≤ 45.7 N authority cap; P_elec ≈220 W peak → ≈45 W terminal.
- **Brake Torque:** tau_brake despin −8…−6 N·m (0–2 s) → capture spike @ 7 s → ≈0 hold.
- **Fuze Logic:** armed step @ 0.85 s; fired single sample at stop step (zoom $x \in [67, 68]$).
- **Body Dynamics:** w_nose → ≈0 after arm; phi_nose constant during hold; $\|q\| \approx 1$.

4. Run Outputs

- **Scopes:** Five distinct scope windows (States, Nav, ZEM, Plant Outputs, Fuze).
- **Figures:** StopFcn white background 3-D trajectory plot, optional live3d animation figure.
- **Workspace:** Data saved to out.simout (Nav 6-state, Timeseries).

5. Monte-Carlo & CEP — Power Generation Reference

Authoritative accuracy pipeline lives in Power Generation (Alternator):

1. Navigate to target repository:
   cd ~/projects/VEL/VEL-repo/pgk_power_sim
2. Execute parallel Monte-Carlo run:
   bash run_mc_parallel.sh (13 workers → worker_k.log + mc_slice_k.mat; single-thread fallback: bash run_mc_v73.sh)
3. Aggregate results in MATLAB:
   Execute mc_aggregate (or combine_mc) to generate pgk_flight_results.mat.
4. Process statistical metrics in MATLAB:
   Execute analyze_results to produce CEP printouts alongside mc_corr_hist.png and mc_vsc_hist.png.
5. **Campaign Values (RESULTS.md §2, v7.5, N = 1300):**

- Mean Authority: 36.8 m
- Residual 95th Percentile ($resid_{95}$): 6.2 m
- Accuracy Metrics: $CEP_{50} \approx 3.0 \text{ m}$ / $CEP_{90} \approx 5.4 \text{ m}$ (Rayleigh-equivalent proxy)
- Minimum Capacitor Voltage (V_{sc} worst): 15.09 V
- Acceptance Criteria: Authority $\geq 3 \times resid_{95} \implies$ **PASS**

6. **Dispersion Profile:** $V_0 \pm 2$, Canard $C_L \pm 20$, Drag ± 10, Crosswind $3 \text{ m/s } (1\sigma)$, Gyro noise 0.005 rad/s.

Caveat (Carry into Report): CEP is a proxy — excludes wind shear, muzzle velocity beyond ± 2, IMU bias drift, and cant error; full-dispersion CEP in the 6-DoF guidance-in-loop Simulink environment is deferred to post-submission hardening.

6. Troubleshooting Quick Reference

Symptom / Error	Root Cause	Resolution
Undefined function 'live3d'	Path missing model directory	Run addpath(<model folder>); execute savepath to persist.
Algebraic loop detected	Incorrect Sensor wiring	Ensure Sensors block has exactly 2 inputs (t, state).
IC / dimension runtime errors	Incorrect workspace mapping	The integrator IC field must be literally set to IC0.
Run terminates prematurely	Incorrect Stop Time setting	Verify Stop Time = 120 (nominal flight duration $\approx 68 \text{ s}$).
Nav km-scale offset vs Truth	Incorrect IMU integration	Ensure Sensors block uses dynamic $dt = t - prev_t$ and includes $+g_vec$ in velocity updates.